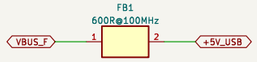
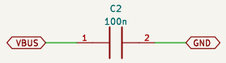
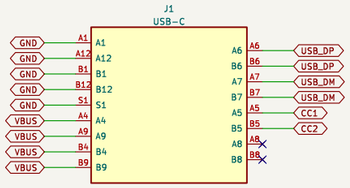
USB-C, protection and enumerated power

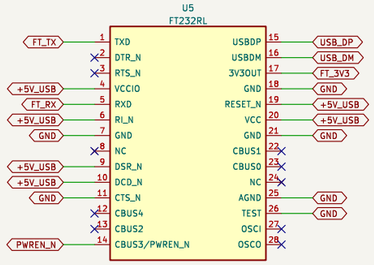


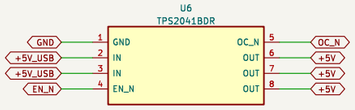


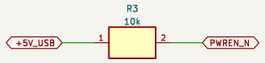


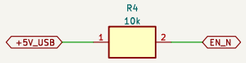


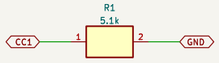




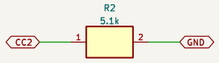


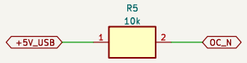


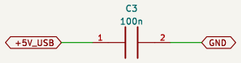


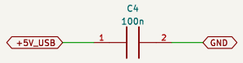




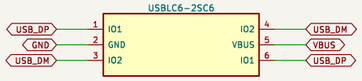


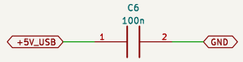




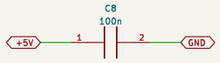


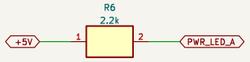


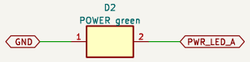












SAB80535-N core, reset, clock and analogue inputs

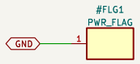
File: sheet2.kicad_sch

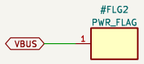
External CODE and XDATA memory

File: sheet3.kicad_sch

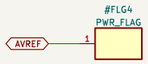
UART isolation, GPIO and test points

File: sheet4.kicad_sch









JP1 OPEN for initial FT_PROG provisioning. Set bus power 500 mA, CBUS3=PWREN#, pull-down on suspend.

Disconnect USB, fit JP1, reconnect. Main rail is off before enumeration and during USB suspend.

No USB-PD. Use a data cable and a powered host/hub. Verify +5V > 4.75 V under load.

F1 = resettable fuse; U6 = current/thermal limited switch. Shield bonded to GND.

USB-C, protection and enumerated power

retro535 – engineering prototype

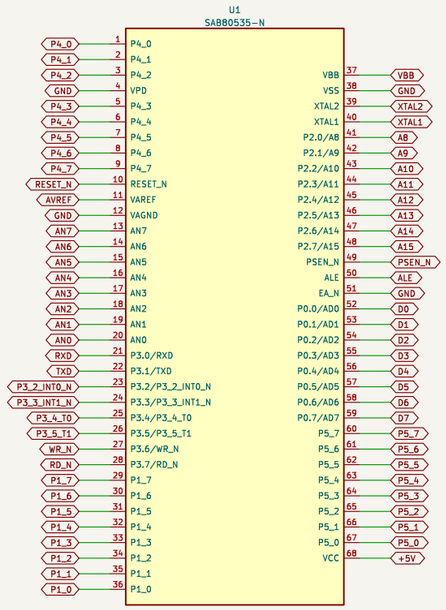
Sheet: /

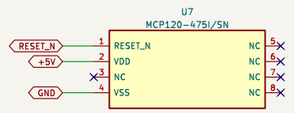
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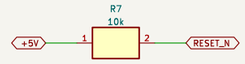
Title: Siemens SAB80535-N Retro Development Board

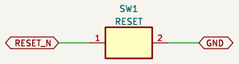
Size: A3	Date: 2026-11-08	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 1/4

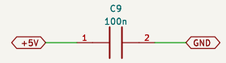
SAB80535–N core, reset, clock and analogue inputs

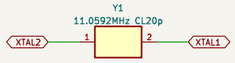






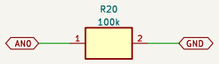


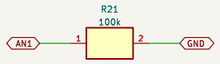


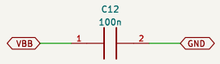


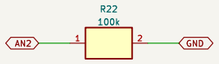


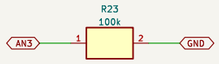


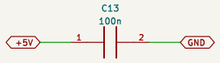




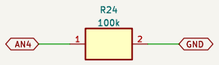


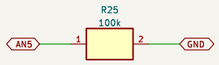




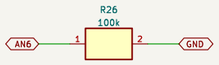


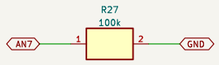


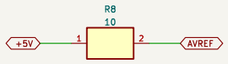


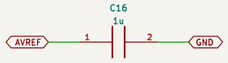












EA# = GND. RESET# is ACTIVE LOW (not standard 8051 polarity). VPD = GND.

VBB pin 37: 100 nF to GND ONLY. Never tie VBB directly to either supply.

11.0592 MHz: 30 pF C0G load capacitors. AN0..AN7: 100k weak pulldowns.

P0 + P2 + P3.6/P3.7 + ALE/PSEN are reserved for memory. See connector sheet.

SAB80535–N core, reset, clock and analogue inputs

retro535 – engineering prototype

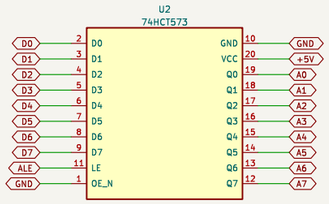
Sheet: /SAB80535–N core, reset, clock and analogue inputs/

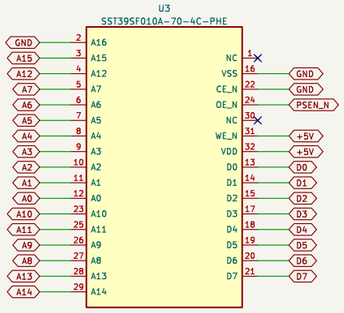
File: sheet2.kicad_sch

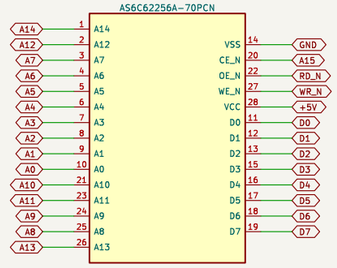
Title: Siemens SAB80535–N Retro Development Board

Size: A3	Date: 2026–11–08	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 2/4

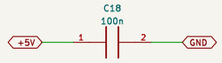
External CODE and XDATA memory



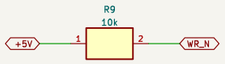


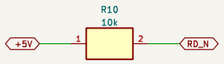












CODE 0000..FFFF = Flash physical 00000..0FFFF. A16=0; upper 64 KiB unused.

Flash /CE=0, /OE=/PSEN, /WE=+5V. REMOVE Flash with ALL POWER OFF for programming.

XDATA 0000..7FFF = 32 KiB SRAM. /CE=A15, /OE=/RD, /WE=/WR. 8000..FFFF unpopulated.

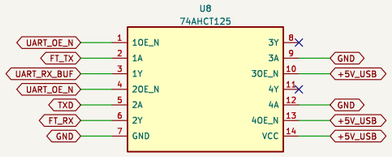
AS6C62256A (VIH=2.2V) is intentional; AS6C62256 without A has VIH=2.4V and zero high margin.

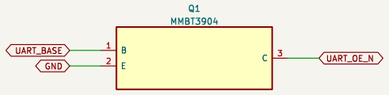
External CODE and XDATA memory
retro535 – engineering prototype
Sheet: /External CODE and XDATA memory/
File: sheet3.kicad_sch

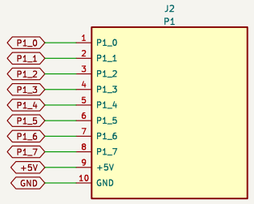
Title: Siemens SAB80535-N Retro Development Board

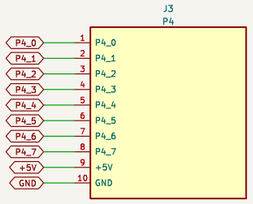
Size: A3	Date: 2026-11-08	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 3/4

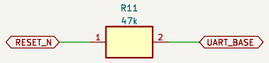
UART isolation, GPIO and test points

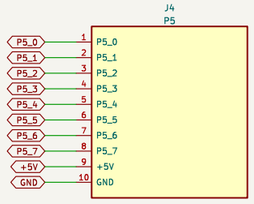


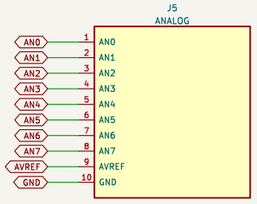


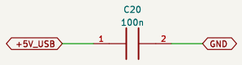


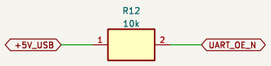


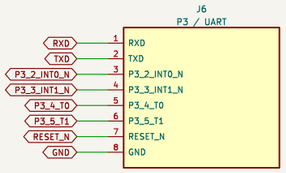


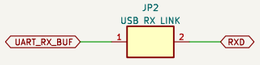


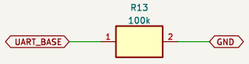


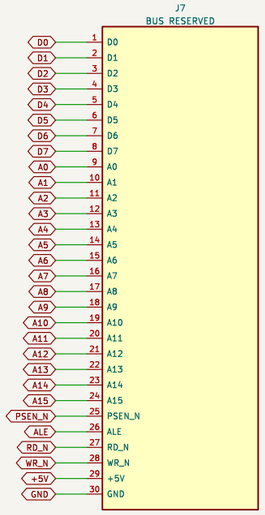


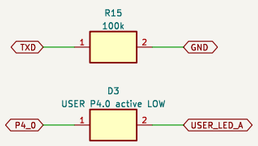


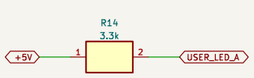


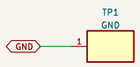


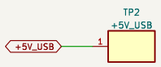


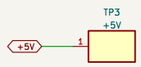


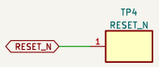


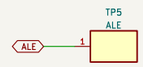


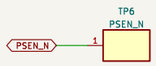


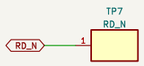


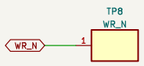


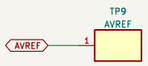


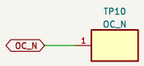












J2/J3/J4: pins 1..8 = port bits 0..7; 9=+5V; 10=GND. 5V TTL, NOT 3.3V safe.

J5: pins 1..8 = AN0..AN7; 9=AVREF output; 10=GND. Do not drive power/reference pins.

J6: 1=RXD, 2=TXD, 3..6=P3.2..P3.5, 7=RESET#, 8=GND. Open JP2 for external RX source.

U8 outputs disabled while RESET# is low. J7 is memory bus: probing only in v1.0.

UART isolation, GPIO and test points

retro535 – engineering prototype

Sheet: /UART isolation, GPIO and test points/

File: sheet4.kicad_sch

Title: Siemens SAB80535-N Retro Development Board

Size: A3	Date: 2026-11-08	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 4/4